

Solar Home Lighting 3

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INTERFACE CONTROL DOCUMENT

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INTERFACE CONTROL DOCUMENT FOR Solar Home Lighting

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1. Overview

This Interface Control Document (ICD) defines how the subsystems of the Smart Solar Home Lighting System will interface with each other. It covers the connections between the solar power, battery, sensors, microcontroller, lighting, camera module, and mobile applications. The ICD outlines electrical, data, and communication interfaces to ensure reliable system integration and operation.

2. References and Definitions

2.1. References

Lighting Global SHS Quality Standards v2.5
Dec 2018

IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces (IEEE 1547a-2020)
2020

National Electrical Code for Solar Photovoltaic Systems (NEC Article 690)
2023

Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources (UL 1741)
2021

ANSI Electric Connectors (ANSI C119.6)
2024

Storage Batteries (NEC Article 480)
2023

2.2. Definitions

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
AC	Alternating Current
DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
kWh	Kilowatts-hour

LED	Light Emitting Diode
USB	Universal Serial Bus
UART	Universal Asynchronous Receiver-Transmitter
BLE	Bluetooth Low Energy
LCD	Liquid Crystal Display
PIR	Passive Infrared Sensor
HTTP	Hypertext Transfer Protocol
MJPEG	Motion JPEG
VGA	Video Graphics Array
IEEE	Institute of Electrical and Electronics Engineers

3. Physical Interface

This section defines the physical characteristics, dimensions, weights, and mounting requirements for all system subsystems. These physical attributes ensure mechanical compatibility during system installation and integration.

3.1. Weight

3.1.1. Physical Controls Subsystem Designed Weight

Component	Weight (g)	Number of Items	Total Weight (g)
ESP32-S3	31	2	62
2.8" TFT LCD	40	1	40
5 Channel Relay Board	90	1	90
PIR Motion Sensor	10	1	10
Reed Switch Door Sensor	10	1	10
Daylight Sensor	17	1	17
LED Recessed Ceiling Light	9.5	3	28.5
LED Flood Security Light	296	2	592
Total Weight (Approx.)			~828 grams

3.1.2. ESP32 Camera and Application Subsystem Designed Weight

Component	Weight (g)	Number of items	Total Weight (g)
Speed Studio XIAO ESP32 S3 Sense	6.6	1	6.6
Total Weight (Approx.)			~6.6 grams

3.1.3 MPPT Charge Controller Designed Weight

Component	Weight (g)	Number of Items	Total Weight (g)
200W Solar Panel	7520	1	7520
12V Lead Acid Battery	10433	1	10433
ESP32	31	1	31
Total Weight (Approx.)			~17,984 grams

3.2. Dimensions

3.2.1. Physical Controls Subsystem Dimensions

Component	Length (in)	Width (in)	Height (in)
ESP32	2.05	1.18	0.59
2.8" TFT LCD	2.8	2	0.3
5 Channel Relay Board	7.17	4.49	0.75
PIR Motion Sensor	1.5	1.5	0.75

Daylight Sensor	2.06	3.0	0.25
LED Recessed Ceiling Light	2.75	2.75	0.62
LED Flood Security Light	4.53	2.76	3.35

3.2.2. ESP32 Camera and Application Subsystem Dimensions

Component	Length (in)	Width (in)	Height (in)
Speed Studio XIAO ESP32 S3 Sense	0.83	0.70	0.59

3.2.3 MPPT Charge Controller Dimensions

Component	Length (in)	Width (in)	Height(in)
200W Solar Panel	42.12	0.38	25.24
12V Lead Acid Battery	6.5	5	7.06
ESP32	2.05	1.18	0.59

3.3. Mounting Locations

3.3.1. Physical Controls Subsystem Panel

The subsystem is intended for indoor mounting near the main residential access point (foyer, hallway, or near a utility closet). Mounting option include:

- Wall mounting: using pre-drilled holes and drywall anchors
- Flush mounting: Inside low-voltage wall box or structured wiring panel

Mounting location must allow access to DC voltage lines and Wi-Fi coverage, and should be within visible and reachable distance for user interaction.

3.3.2. Physical Controls Subsystem Sensors

- **Motion Sensor**
 - The motion sensor will be mounted below the ESP32-EYE camera, allowing the 30 feet of motion detection range to be utilized for camera triggering purposes.
- **Reed Switch Door Sensor**
 - This will be mounted at the top of the door frame to indicate door status.
- **Daylight Sensor**

- This will be mounted to the roof alongside the solar panel to get an accurate light level readout.

3.3.3. ESP32 Camera and Application Subsystem

- **ESP32-Camera**

- Indoor facing outwards in wall mounted casing. In outdoor settings, the esp32-camera will be placed on a wall mount in weather-resistant housing with transparent lens cover.

3.3.4 MPPT Charge Controller

- **12V battery and charge controller**

- Must be in proximity to one another, preferably in the same location.
- Indoors or safe from harsh weather elements, like a cabinet or storage house.

- **Solar Panel**

- Must be on an elevated outdoor structure with full sun exposure.
- Securely mounted for windy conditions.

4. Thermal Interface

This section defines the thermal characteristics, expected heat dissipation, and required cooling provisions for each subsystem. All thermal values are design estimates based on component datasheets and intended operating conditions.

4.1. Physical Controls Subsystem

The Physical Controls subsystem generates minimal thermal load during operation. Passive cooling through ventilation slots in the enclosure is sufficient under standard indoor ambient temperatures.

- Cooling method: passive ventilation
- A copper thermal pad beneath LTC3119 and LT8610AC is required (datasheet guidance).

The enclosure material is ABS plastic, suitable for residential installations. The operating temperature range is between 0-70°C.

4.1.1. Physical Controls Thermal Parameters

Component	Estimated Power Dissipation (W)	θ_{JA} (°C/W)	Expected Junction Rise (°C)
LTC3119 Buck-Boost (12 V Conditioner)	1.0–1.2 W	40 °C/W	40–48 °C

LT8610AC 3.3 V Regulator	0.8–1.0 W	45 °C/W	36–45 °C
ESP32-S3 (Relay MCU)	~0.8 W	55 °C/W	~44 °C
ESP32-S3 (Touchscreen HMI)	~0.6 W	55 °C/W	~33 °C
Relay Drivers (2N2222)	<0.2 W each	70 °C/W	<14 °C
TFT LCD Backlight	~0.5 W	-	<10 °C
Sensors (PIR, reed, daylight)	<0.05 W	-	negligible

4.2. ESP32 Camera and Application Subsystem

Typical operating temperature for a Speed Studio XIAO ESP32 S3 Sense is between 32°F – 122°F (0°C – 50°C). The heat dissipation is minimal (<1 W power draw) but airflow will be utilized for enclosed environments

4.2.1. Camera and Application Thermal Parameters

4.3. MPPT Charge Controller Subsystem

No active cooling is required as passive cooling is sufficient under standard indoor ambient conditions, with heatsinking applied to MOSFETs and inductors. Slotted ventilation can assist in cooling. Thermal isolation is maintained between heat-generating components and temperature-sensitive sensors.

4.3.1. MPPT Charge Controller Thermal Parameters

Component	Estimated Power Dissipation (W)	θ_{JA} (°C/W)	Expected Junction Rise (°C)
LM5176 Buck-Boost Controller	~1.2–1.5 W	~40 °C/W	~48–60 °C
LM3151MHX-3.3 Buck Regulator	~0.8–1.0 W	~45 °C/W	~36–45 °C
ESP32-WROOM-32D MCU	~0.6–0.8 W	~55 °C/W	~33–44 °C

MOSFETs (CSD17310Q5A or similar)	~1.0–2.0 W	~35 °C/W	~35–70 °C
ACS781LL Hall Sensors	~0.2-0.3 W	~60 °C/W	~12–18 °C
NTC Thermistor (10k B3950)	<0.05 W	-	Negligible
Resistor Networks (Feedback/Dividers)	<0.1 W	-	Negligible
Terminal Block / Connectors (MC4, PCB)	<0.2 W	-	Negligible

5. Electrical Interface

This section defines the electrical power interfaces, connector pinouts, voltage levels, current limits, signal characteristics, and communication protocols between all subsystems.

5.1. Primary Input Power

5.1.1. Physical Controls Subsystem

The physical control will be powered by the DC battery bank from the charge controller subsystem.

- Input Voltage Range: +10.8 VDC to +14.2 VDC (nominal)
- Input Type: 2-pin terminal block (fused type)

5.1.2. ESP32 Camera and Application Subsystem

Supply voltage will be 3.3 V regulated via USB-C or battery pack. Typical current draw will be 160-200 mA during streaming, with peak current up to 500 mA.

5.1.3. MPPT Charge Controller Subsystem

The primary input power for both the battery and the charge controller will be directly from the Solar Panel.

5.2. Voltage and Current Ratings

5.2.1. Physical Controls Subsystem Internal Power Consumption

Component	Input Voltage (VDC)	Current (mA)	Power (W)
ESP32 Microcontrollers	3.3	500	1.65
TFT LCD Touch Controller	3.3	150	0.495
Relay Control Board	3.3	150	0.495
Voltage Regulator (3.3V)	12 (nominal)	75	0.9
Voltage Regulator (12V)	12 (nominal)	90	1.1
PIR Motion Sensor	3.3	3	0.009
Reed Switch Door Sensor	3.3	<1	N/A
Daylight Sensor	3.3	2	0.007
Internal Subtotal			~4.7 W

5.2.2. Physical Controls Subsystem Lighting Output Load

Load Type	Quantity	Power per Unit (W)	Total Load (W)
LED Recessed Ceiling Light	3	2.5	7.5
LED Flood Security Light	2	10	20
Total Peak Load			27.5 W

5.2.3. Camera Subsystem and Application

Component	Input Voltage	Typical Current	Peak Current	Power
XIAO ESP32-S3	3.3 V	180 mA	300 mA	0.59 W

5.2.4 MPPT Charge Controller

Load Type	Input Voltage (VDC)	Current (mA)	Power (W)
200W Solar Panel	18.8	10.6	200
12V Lead Acid Battery	~14V	100 mAh	
ESP32	3.3	250	0.825
Total Peak Load			~201 W

5.3. Polarity Reversal

To protect against reverse polarity connection on the 12VDC input, the physical control subsystem includes an in-line Schottky diode for polarity protections. This diode will be rated for a current exceeding the max peak load continuous forward current, providing reliable protection for the subsystems peak load. Under normal load conditions the diode permits current flow with a typical voltage drop of approximately 0.5V, resulting in minimal power loss.

5.4. Signal Interfaces

This section defines the signal-level interfaces used by each subsystem, including voltage levels, expected current, and direction. All unused ESP32 pins are intentionally omitted.

5.4.1. Physical Controls Subsystem Analog and Digital Signals

Signal Name	ESP32 Pin	Direction	Voltage	Max Current	Protocol / Function
UART2_RX	IO17	Input	3.3 V	—	UART receive, 115200 baud, 8-N-1
UART2_TX	IO16	Output	3.3 V	—	UART transmit, 115200 baud,

					8-N-1
PIR_IN	IO14	Input	3.3 V	<5 mA	Digital motion detection
REED_IN	IO36	Input	3.3 V	<1 mA	Door open/closed detection
IR_SENSOR_IN	IO10	Input	3.3 V	<5 mA	Auxiliary IR sensor
SPARE_SENSOR_IN	IO11	Input	3.3 V	<5 mA	Reserved future digital sensor
DAYLIGHT_ADC	IO21	Input	0–3.3 V	—	12-bit ADC daylight measurement
RELAY1_CTRL	IO12	Output	3.3 V	Logic only	Controls Relay 1 (via 2N2222 driver)
RELAY2_CTRL	IO19	Output	3.3 V	Logic only	Controls Relay 2
RELAY3_CTRL	IO9	Output	3.3 V	Logic only	Controls Relay 3
RELAY4_CTRL	IO13	Output	3.3 V	Logic only	Controls Relay 4
RELAY5_CTRL	IO35	Output	3.3 V	Logic only	Controls Relay 5
POWER_LED	IO2	Output	3.3 V	<20 mA	Status/power indication LED
TOUCH_SDA	IO21	Bidirectional	3.3 V	—	I ² C data, 400 kHz
TOUCH_SCL	IO11	Output	3.3 V	—	I ² C clock, 400 kHz
TOUCH_IRQ	IO9	Input	3.3 V	<5 mA	Touch interrupt (active-low)
TFT_SCK	IO18	Output	3.3 V	—	SPI clock
TFT_MOSI	IO20	Output	3.3 V	—	SPI MOSI
TFT_MISO	IO19	Input	3.3 V	—	SPI MISO
TFT_CS	IO5	Output	3.3 V	—	SPI chip select
TFT_DC	IO2	Output	3.3 V	—	Data/command select
TFT_RST	IO4	Output	3.3 V	—	Reset line
TFT_BL_PWM	IO14	Output	3.3 V	—	Backlight PWM, 1 kHz

The Physical Controls Subsystem exchanges digital, analog, and communication signals required for sensing, relay activation, and interaction with the touchscreen interface. This section defines each electrical signal used by the subsystem, including direction, voltage

level, expected current, and protocol characteristics. Only active signals are listed; all unused ESP32 pins have been intentionally omitted. The interface enables coordination between sensors, user controls, and lighting outputs while ensuring compatibility with the system's power and communication requirements.

5.4.2. MPPT Charge Controller Analog and Digital Signals

Signal Name	ESP32 Pin	Direction	Voltage	Max Current	Protocol / Function
RTS# to EN	EN	Input	3.3V	-	Reset via UART RTS line
Thermistor	GPIO2	Input	0–3.3 V	-	NTC thermistor measures battery temperature
Load Current	GPIO4	Input	0–3.3 V	-	ACS781LL output for load current
Battery Voltage	GPIO34	Input	0–3.3 V	-	Voltage divider from solar panel to ADC
Battery Current	GPIO35	Input	0–3.3 V	-	ACS781LL output for battery current
UART_TXD	TXD0	Input	3.3V	-	UART transmit to charge controller
UART_RXD	RXD0	Output	3.3V	-	UART receive from charge controller
3.3 VIN	3V3	Input	3.3V	-	ESP32 supply voltage
DTR# to IO0	GPIO0	Input	3.3V	-	Boot mode via UART DTR line
MPPT_DAC	GPIO25	Output	0–3.3 V	~12mA	DAC output to LM5176 FB pin for MPPT control
Load Switch	GPIO26	Output	3.3V	~12mA	Disconnects the load from the battery
Panel Voltage	GPIO32	Input	0–3.3 V	-	Voltage divider from solar panel to ADC
Panel Current	GPIO33	Input	0–3.3 V	-	ACS781LL output for panel current

5.4.3. MPPT Charge Controller Communication Interface

The communication interface of the charge controller is centered on the ESP32. Using UART, DAC/PWM, and ADC, to monitor and supply an efficient charge to the battery. UART provides a wired interface for debugging and inter-microcontroller messaging, with TXD and RXD lines operating at 115200 baud and 3.3 V logic levels. The ESP32 generates either a DAC output or PWM signal to the LM5176 feedback pin, enabling real-time duty cycle adjustment for maximum power point tracking. Multiple ADC channels measure solar panel voltage and current, battery voltage and current, load current, and battery temperature, all scaled to 0–3.3 V inputs. All communication signals are referenced to the 3.3 V supply rail, ensuring consistent logic levels across analog and digital interfaces.

6. Communications / Device Interface Protocols

6.1. Physical Controls Subsystem Communication Interface

The Physical Controls Subsystem uses UART, I²C, SPI, and PWM signaling standards for communication and device control. UART is used for inter-microcontroller messaging between the HMI and relay controller. The capacitive touch controller communicates using an I²C Fast-Mode interface at 400 kHz. The TFT display operates over a dedicated SPI bus. A PWM signal is used to regulate the backlight intensity of the display. All communication signaling operates at 3.3 V logic levels.

6.1.1. Physical Controls Communication Parameters

Interface	ESP32 Pins (Relay MCU)	Voltage	Direction	Protocol Parameters	Description
UART2 (Inter-ESP32 Link)	TX: IO16, RX: IO17	3.3 V	Bidirectional	115200 baud, 8-N-1	Command and status messaging between Relay MCU and HMI MCU
I ² C (Touchscreen FT6206)	SDA: IO21, SCL: IO11	3.3 V	Bidirectional	400 kHz Fast Mode	Capacitive touch event and coordinate reporting
PWM (Backlight Control)	IO14	3.3 V	Output	1 kHz, 0–100% duty	TFT LCD backlight control (if physically integrated)

6.2. Wireless Communications (WiFi)

Standard: IEEE 80.2.11 b/g/n at 2.4 GHz

Function: Provides primary communication path between the ESP32 camera subsystem and Android mobile application

Bandwidth: MJPEG video stream operates at up to 2 Mbps depending on resolution. MQTT sensor/control messages operate under 10 kbps, which ensures minimal overhead

6.3. Host Device

Mobile Device: Android smartphone or tablet running on Flutter/SwiftUI based application.

6.4. Video Interface

Type: MJPEG over HTTP streaming from an onboard ESP32 server

Protocol: HTTP GET request handled by ESP32 web server library

Resolution: VGA (640x480) 15 fps